```
$ aws dynamodb update-table --table-name todo-task \                    ①
➥ --attribute-definitions AttributeName=uid,AttributeType=S \
➥ AttributeName=tid,AttributeType=N \
➥ AttributeName=category,AttributeType=S \                             ②
➥ --global-secondary-index-updates '[{\
➥ "Create": {\                                                         ③
➥ "IndexName": "category-index", \
➥ "KeySchema": [{"AttributeName": "category",
➥ "KeyType": "HASH"}, \                                                ④
➥ {"AttributeName": "tid", "KeyType": "RANGE"}], \
➥ "Projection": {"ProjectionType": "ALL"}, \                           ⑤
➥ "ProvisionedThroughput": {"ReadCapacityUnits": 5, \
➥ "WriteCapacityUnits": 5}\
➥ }}]'
```

① Adds a global secondary index by updating the table that has already been created previously

② Adds a category attribute, because the attribute will be used in the index

③ Creates a new secondary index

④ The category attribute is the partition key, and the tid attribute is the sort key.

⑤ All attributes are projected into the index.

Creating a global secondary index takes about five minutes. You can use the CLI to find out if the index is ready like this:

```
$ aws dynamodb describe-table --table-name=todo-task \
➥ --query "Table.GlobalSecondaryIndexes"
```

The next listing shows the code to query the global secondary index to fetch tasks by category with the help of the `task-la` command.

Listing 12.12 nodetodo: Retrieving tasks from a global secondary index (index.js)

```
if (input['task-la'] === true) {
  const yyyymmdd = moment().format('YYYYMMDD');
  const params = {
    KeyConditionExpression: 'category = :category',                     ①
    ExpressionAttributeValues: {
      ':category': {S: input['<category>']}
```